

Pooled MV-STGNN & Baselines

Core Setup & Protocol

Scope: 20 subjects pooled, 5 movement classes (Rest excluded), tested under a 5 fold cross-repetition split (evaluating generalization to new repetitions for known subjects, not cross-subject LOSO).

Leakage-Free Processing: Per-subject Z-score normalization computed strictly on training repetitions; filtering uses fixed coefficients.

Proposed Model: MV-STGNN (~189K params)

Multi-View Graph: Combines anatomical geometry, subject-specific coherence, dynamic attention, and learned adjacency matrices.

Subject Adaptivity: Uses GroupNorm and Subject FiLM conditioning to adapt to individual signal profiles within a single model.

Architecture: Temporal stem (TCN) → FiLM → Spatio-temporal graph blocks → Synergy pooling → Dual attention readout.

Fair Baseline Comparison

To ensure statistically valid head-to-head evaluation, all deep baselines match MV-STGNN's training budget, Subject FiLM conditioning, loss functions, and parameter scale (~0.15M–0.25M):

- **Random Forest:** ~480 hand-crafted features per window (time, spectral, Hjorth, AR).
- **1D-CNN (~0.15M):** 3 convolutional blocks with GroupNorm and global average pooling.
- **BiLSTM (~0.20M):** 2-layer BiLSTM over 40 temporal patch tokens.
- **Transformer (~0.18M):** 3-layer pre-LN encoder with multi-head attention.